RAJALAKSHMI ENGINEERING COLLEGE

Monish A 251001155-2025-IT

M2

Started on	Sunday, 30 August 2026, 9:13 PM
State	Finished
Completed on	Sunday, 30 August 2026, 9:13 PM
Time taken	28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int found = 0;

void twoSumRec(int arr[], int low, int high, int x) {
    if (low >= high)
        return;

    int sum = arr[low] + arr[high];

    if (sum == x) {
        printf("%d\n", arr[low]);
        printf("%d\n", arr[high]);
        found = 1;
        return;
    } else if (sum < x) {
        twoSumRec(arr, low + 1, high, x);
    } else {
        twoSumRec(arr, low, high - 1, x);
    }
}

int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    int x;
    scanf("%d", &x);

    twoSumRec(arr, 0, n - 1, x);

    if (!found)
        printf("No\n");

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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